TRIPATHI STUDY ZONE

JEE MAIN | NEET | AIIMS | KVPY | BITSAT

Mock Test 7

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

Section - A

1. Using mass(M), length(L), time(T) and electric current (A) as fundamental quantities the dimensions of permittivity will be

   (1) $MLT^{-1}A^{-1}$ (2) $MLT^{-2}A^{-2}$

   (3) $M^{-1}L^{-3}T^{4}A^{2}$ (4) $M^{2}L^{-2}T^{-2}A^{2}$

2. $y = 2\,(\text{cm})\sin\left[\frac{\pi t}{2} + \phi\right]$

   What is the maximum acceleration of the particle executing the SHM?

   (1) $\frac{\pi}{2}$ cm/s^2 (2) $\frac{\pi^2}{2}$ cm/s^2

   (3) $\frac{\pi^2}{4}$ cm/s^2 (4) $\frac{\pi}{4}$ cm/s^2

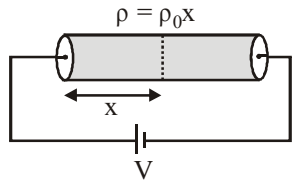
3. A coil having N turns is wound tightly in the form of a spiral with inner and outer radii a and b respectively. When a current I passes through the coil, the magnetic field at the center is

   (1) $\frac{\mu_o NI}{b}$ (2) $\frac{2\mu_o NI}{a}$

   (3) $\frac{\mu_0 NI}{2(b-a)}\ln\frac{b}{a}$ (4) $\frac{\mu_0 IN}{2(b-a)}\ln\frac{a}{b}$

Space for Rough Work

4. A cylindrical solid of length L and radius a having varying resistivity given by $\rho = \rho_0 x$, where ρ_0 is a positive constant and x is measured from left end of solid. The cell shown in the figure having emf V and negligible internal resistance. The electric field as a function of x is best described by



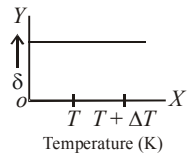
(1) $\frac{2V}{L^2}x$ (2) $\frac{2V}{\rho_0 L^2}x$

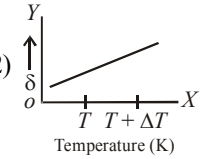
(3) $\frac{V}{L^2}x$ (4) None of these

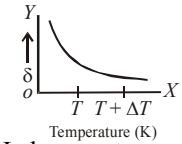
5. A mass *m* hangs with the help of a string wrapped around a pulley on a frictionless bearing. The pulley has mass *m* and radius *R*. Assuming pulley to be a perfect uniform circular disc, the acceleration of the mass *m*, if the string does not slip on the pulley, is:

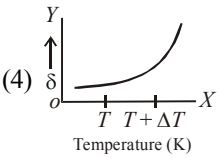
(1) g (2) $\frac{2}{3}g$ (3) $\frac{g}{3}$ (4) $\frac{3}{2}g$

6. An ideal gas is initially at temperature *T* and volume *V*. Its volume is increased by ΔV due to an increase in temperature ΔT, pressure remaining constant. The quantity $\delta = \frac{\Delta V}{V\Delta T}$ varies with temperature as

(1)


(2)


(3)


(4)


7. A Hydrogen atom and a Li^{++} ion are both in the second excited state. If ℓ_H and ℓ_{Li} are their respective electronic angular momenta, and E_H and E_{Li} their respective energies, then

(1) $\ell_H > \ell_{Li}$ and $|E_H| > |E_{Li}|$

(2) $\ell_H = \ell_{Li}$ and $|E_H| < |E_{Li}|$

(3) $\ell_H = \ell_{Li}$ and $|E_H| > |E_{Li}|$

(4) $\ell_H < \ell_{Li}$ and $|E_H| < |E_{Li}|$

8. The temperature of reservoir of Carnot's engine operating with an efficency of 70% is 1000 kelvin. The temperature of its sink is

(1) 300 K (2) 400 K

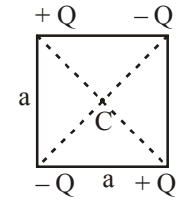
(3) 500 K (4) 700 K

9. A radar has a power of 1kW and is operating at a frequency of 10 GHz. It is located on a mountain top of height 500 m. The maximum distance upto which it can detect object located on the surface of the earth (Radius of earth = 6.4×10^6m) is :

(1) 80 km (2) 16 km

(3) 40 km (4) 64 km

10. What is the electric potential at the centre of the square?

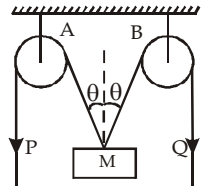


(1) zero (2) $kq/a\sqrt{2}$

(3) kq/a^2 (4) None of these

Space for Rough Work

11. In the arrangement shown in the Fig, the ends *P* and *Q* of an unstretchable string move downwards with uniform speed *U*. Pulleys *A* and *B* are fixed. Mass *M* moves upwards with a speed



(1) $2U\cos\theta$

(2) $U/\cos\theta$

(3) $2U/\cos\theta$

(4) $U\cos\theta$

12. In Young's double slit experiment, the slits are illuminated by white light. The distance between two slits is d and screen is D distance apart from the slits. Some wavelengths are missing on the screen in front of one of the slits. These wavelengths are

(1) $\lambda = \frac{d^2}{D}$ (2) $\lambda = \frac{2d^2}{D}$

(3) $\lambda = \frac{d^2}{2D}$ (4) $\lambda = \frac{2d^2}{3D}$

13. A rectangular block of mass m and area of cross-section A floats in a liquid of density ρ. If it is given a small vertical displacement from equilibrium it undergoes oscillation with a time period T. Then

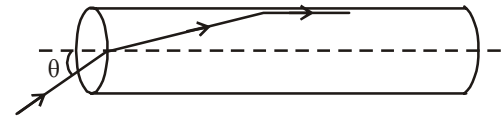
(1) $T \propto \frac{1}{\sqrt{A}}$ (2) $T \propto \frac{1}{\rho}$

(3) $T \propto \frac{1}{\sqrt{m}}$ (4) $T \propto \sqrt{\rho}$

14. There are two wires of the same length. The diameter of second wire is twice that of the first. On applying the same load to both the wires, the extension produced in them will be in ratio of

(1) 1 : 4 (2) 1 : 2 (3) 2 : 1 (4) 4 : 1

15. A transparent solid cylindrical rod has a refractive index of $\frac{2}{\sqrt{3}}$. It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure.

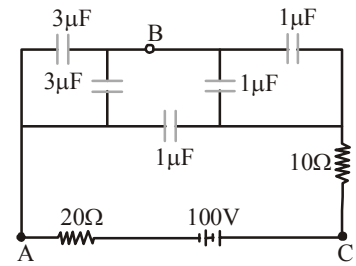


The incident angle θ for which the light ray grazes along the wall of the rod is

(1) $\sin^{-1}\left(\sqrt{3}/2\right)$ (2) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(3) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (4) $\sin^{-1}\left(1/2\right)$

16. In the figure below, what is the potential difference between the point A and B and between B and C respectively in steady state



(1) $V_{AB} = V_{BC} = 100\,V$

(2) $V_{AB} = 75, V_{BC} = 25\,V$

(3) $V_{AB} = 25V, V_{BC} = 75V$

(4) $V_{AB} = V_{BC} = 50V$

Space for Rough Work

17. A block is placed on a frictionless horizontal table. The mass of the block is m and springs are attached on either side with force constants K_1 and K_2. If the block is displaced a little and left to oscillate, then the angular frequency of oscillation will be

(1) $\left(\frac{K_1+K_2}{m}\right)^{\frac{1}{2}}$ (2) $\left[\frac{K_1K_2}{m(K_1+K_2)}\right]^{\frac{1}{2}}$

(3) $\left[\frac{K_1K_2}{(K_1-K_2)m}\right]^{\frac{1}{2}}$ (4) $\left[\frac{K_1^2+K_2^2}{(K_1+K_2)m}\right]^{\frac{1}{2}}$

18. A bullet looses $\left(\frac{1}{n}\right)^{th}$ of its velocity passing through one plank. The number of such planks that are required to stop the bullet can be:

(1) $\frac{n^2}{2n-1}$ (2) $\frac{2n^2}{n-1}$

(3) infinite (4) n

19. An ideal coil of 10H is connected in series with a resistance of 5Ω and a battery of 5V. 2second after the connection is made, the current flowing in ampere in the circuit is

(1) $(1-e^{-1})$ (2) $(1-e)$

(3) e (4) e^{-1}

20. A spacecraft of mass 'M', moving with velocity 'v' suddenly breaks into two pieces. After the explosion mass 'm' becomes stationary. What is the velocity of the other part of the spacecraft?

(1) $\frac{Mv}{M-m}$ (2) v

(3) $\frac{mv}{M}$ (4) $\frac{M-m}{m}v$

Section - B

21. In a series LCR circuit $R = 200\Omega$ and the voltage and the frequency of the main supply is 220V and 50 Hz respectively. On taking out the capacitance from the circuit the current lags behind the voltage by 30°. On taking out the inductor from the circuit the current leads the voltage by 30°. The power dissipated in the LCR circuit is _____ watt.

22. In an isothermal expansion of 10g of gas from volume V to 2V the work done by the gas is 575J. What is the root mean square speed (in m/s) of the molecules of the gas at that temperature?

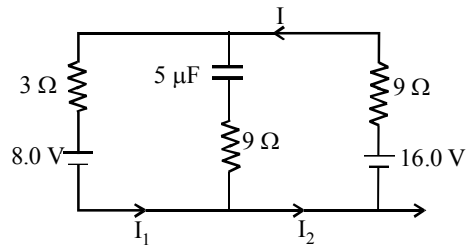
23. An automobile travelling with a speed of 60 km/h, can brake to stop within a distance of 20m. If the car is going twice as fast i.e., 120 km/h, the stopping distance (in metre) will be

24. At the centre of a circular current carrying coil of radius 5 cm magnetic field due to earth is 0.5×10^{-5} W/m^2. What should be the current (in A) flowing through the coil so that it annuls the earth's magnetic field?

25. An air bubble of radius 0.1 cm is in a liquid having surface tension 0.06 N/m and density 10^3 kg/m^3. The pressure inside the bubble is 1100 Nm^{-2} greater than the atmospheric pressure. At what depth (in metre) is the bubble below the surface of the liquid? ($g = 9.8$ ms^{-2})

26. The escape velocity for a body of mass 1 kg from the earth surface is 11.2 kms^{-1}. The escape velocity for a body of mass 100 kg would be _____ kms^{-1}.

27. The circuit shown here has two batteries of 8.0 V and 16.0 V and three resistors 3 Ω, 9 Ω and 9 Ω and a capacitor of 5.0 μF.



How much is the current (I) (in A) in the circuit in steady state?

28. A beam of light of intensity 12 watt/cm^2 is incident on a totally reflecting plane mirror of area 1.5 cm^2, then the force (in newton) acting on the mirror will be

29. A liquid drop having 6 excess electrons is kept stationary under a uniform electric field of 25.5 kVm^{-1}. The density of liquid is 1.26×10^3 kg m^{-3}. The radius of the drop is _____ metre. (neglect buoyancy).

30. Two waves of wavelengths 99 cm and 100 cm both travelling with velocity 396 m/s are made to interfere. The number of beats produced per second is

CHEMISTRY

Section - A

31. Ethylene dichloride and ethylidine chloride are isomeric compounds. The false statement about these isomers is that they
    (1) react with alcoholic potash and give the same product.
    (2) are position isomers.
    (3) contain the same percentage of chlorine.
    (4) are both hydrolysed to the same product.

32. By what ratio the average velocity of the molecule in gas changes when the temperature is raised from 50 to 200 ºC ?
    (1) $\frac{1.21}{1}$ (2) $\frac{1.46}{1}$
    (3) $\frac{2}{1}$ (4) $\frac{4}{1}$

33. Which of the following would not give 2-phenylbutane as the major product in a Friedel-Crafts alkylation reaction ?
    (1) 1-butene + HF
    (2) 2-butanol + H_2SO_4
    (3) Butanoyl chloride + $AlCl_3$ then Zn, HCl
    (4) Butyl chloride + $AlCl_3$

34. Moissan boron is
    (1) amorphous boron of ultra purity
    (2) crystalline boron of ultra purity
    (3) amorphous boron of low purity
    (4) crystalline boron of low purity

35. Which reaction characteristics are changed by the addition of a catalyst to a reaction at constant temperature ?
    (i) Activation energy
    (ii) Equilibrium constant
    (iii) Reaction enthalpy
    (1) (i) only (2) (iii) only
    (3) (i) and (ii) only (4) all of these

36. A metal M reacts with N_2 to give a compound 'A' (M_3N). 'A' on heating at high temperature gives back 'M' and 'A' on reacting with H_2O gives a gas 'B'. 'B' turns $CuSO_4$ solution blue on passing through it. M and B can be :
(1) Al & NH_3 (2) Li & NH_3
(3) Na & NH_3 (4) Mg & NH_3

37. Asthma patient use a mixture of .... for respiration.
(1) O_2 and CO_2 (2) O_2 and He
(3) O_2 and NH_3 (4) O_2 and CO

38. When ethanal reacts with CH_3MgBr and C_2H_5OH/dry HCl, the product formed are :
(1) ethyl alcohol and 2-propanol
(2) ethane and hemi acetal
(3) 2-propanol and acetal
(4) propane and methyl acetate

39. Amongst the following salts of iron, which is most unstable in aqueous solutions?
(1) $K_3[Fe(CN)_6]$ (2) $Fe_2(SO_4).9H_2O$
(3) $FeSO_4.7H_2O$ (4) FeI_3

40. Given the molecular formula of the hexa-coordinated complexes (i) $CoCl_3.6NH_3$, (ii) $CoCl_3.5NH_3$, (iii) $CoCl_3.4NH_3$
If the number of co-ordinated NH_3 molecules in i, ii and iii respectively are 6, 5, 4, the primary valencies in (i), (ii) and (iii) are :
(1) 6, 5, 4 (2) 3, 2, 1
(3) 0, 1, 2 (4) 3, 3, 3

41. Polyethylene is
(1) Random copolymer
(2) Homopolymer
(3) Alternate copolymer
(4) Crosslinked copolymer

42. The reason for "drug induced poisoning" is :
(1) Binding reversibly at the active site of the enzyme
(2) Bringing conformational change in the binding site of enzyme
(3) Binding irreversibly to the active site of the enzyme
(4) Binding at the allosteric sites of the enzyme

43. With a change in hybridization of the carbon bearing the charge, the stability of a carbanion increases in the order :
(1) $sp < sp^2 < sp^3$ (2) $sp < sp^3 < sp^2$
(3) $sp^3 < sp^2 < sp$ (4) $sp^2 < sp < sp^3$

44. Which of the following compounds are not arranged in order of decreasing reactivity towards electrophilic substitution?
(1) Methoxybenzene > Toluene > Bromobenzene
(2) Phenol > N-Propylbenzene > Benzoic acid
(3) Chlorotoluene > Para Nitro toluene > 2-chloro-4-Nitrotoluene
(4) Benzoic acid > Phenol > N-Propylbenzene

45. Which of the given sets of temperature and pressure will cause a gas to exhibit the greatest deviation from ideal gas behaviour ?
(1) 100 ºC & 4 atm (2) 100 ºC & 2 atm
(3) –100 ºC & 4 atm (4) 0 ºC & 2 atm

46. Which is not true for beryllium ?
(1) Beryllium is amphoteric.
(2) It forms unusual carbide, Be_2C.
(3) $Be(OH)_2$ is basic.
(4) Beryllium halides are electron deficient.

47. Which of the following pairs show reverse properties on moving along a period from left to right and from up to down in a group :
(1) Nuclear charge and electron affinity
(2) Ionization energy and electron affinity
(3) Atomic radius and electron affinity
(4) None of these

48. An organic amino compound reacts with aqueous nitrous acid at low temperature to produce an oily nitrosoamine. The compound is:
(1) CH_3NH_2 (2) $CH_3CH_2NH_2$
(3) $CH_3CH_2NH.CH_2CH_3$ (4) $(CH_3CH_2)_3N$

49. A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the solution, a white precipitate is obtained which does not dissolve in dil. HNO_3. The anion could be
(1) CO_3^{2-} (2) Cl^-
(3) SO_4^{2-} (4) S^{2-}

50. Energy of activation is lowest for which reaction?
(1) $RCH_2\overset{+}{O}H_2 \rightarrow R\overset{+}{C}H_2$
(2) $R_2CH\overset{+}{O}H_2 \rightarrow R_2\overset{+}{C}H$
(3) $R_3C\overset{+}{O}H_2 \rightarrow R_3C^+$
(4) All have same E_{act}

Section - B

51. The pK_a of HCOOH is 3.8 and pK_b of NH_3 is 4.8, find the pH of aqueous solution of 1M $HCOONH_4$.

52. The number of electrons passing per second through a cross-section of copper wire carrying 10^{-6} amperes of current is found to be $x \times 10^2$. Find the value of x.

53. Number H-atoms present in 0.046 g of ethanol is $x \times 10^{21}$. Find the value of x.

54. Calculate the radius (in pm) of the largest sphere which fits properly at the centre of the edge of a body centred cubic unit cell. (Given edge length is 100 pm)

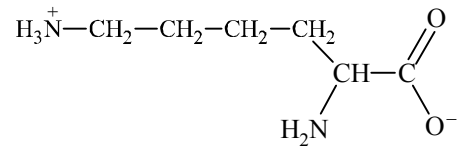
55. What is the molarity of H_2SO_4 solution if 25mL is exactly neutralized with 32.63 mL of 0.164 M, NaOH ?

56. 1.500g of hydroxide of a metal gave 1.000g of its oxide on heating. What is the equivalent mass of the metal ?

57. The enthalpy change on freezing of 1 mole of water at 5 °C to ice at –5 °C is :
[Given $\Delta_{fus}H = 6$ kJ mol^{-1} at 0 °C,
$C_p(H_2O, l) = 75.3$ J mol^{-1} K^{-1},
$C_p(H_2O, s) = 36.8$ J mol^{-1} K^{-1}]

58. The ground state energy of hydrogen atom is –13.6 eV. Calculate the energy of second excited state of He^+ ion in eV.

59. The total number of basic groups in the following form of lysine is



60. What quantity (in mL) of a 45% acid solution of a mono-protic strong acid must be mixed with a 20% solution of the same acid to produce 800 mL of a 29.875% acid solution?

MATHEMATICS

Section - A

61. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in

(1) $\left(0, \frac{\pi}{2}\right)$ (2) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(3) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (4) $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$

62. A tangent to the hyperbola $\frac{x^2}{4} - \frac{y^2}{2} = 1$ meets x-axis at P and y-axis at Q. Lines PR and QR are drawn such that OPRQ is a rectangle (where O is the origin). Then R lies on :

(1) $\frac{4}{x^2} + \frac{2}{y^2} = 1$ (2) $\frac{2}{x^2} - \frac{4}{y^2} = 1$

(3) $\frac{2}{x^2} + \frac{4}{y^2} = 1$ (4) $\frac{4}{x^2} - \frac{2}{y^2} = 1$

63. With the usual notation $\int_1^2 \left([x^2] - [x]^2 \right) dx$ is equal to

(1) $4 + \sqrt{2} - \sqrt{3}$ (2) $4 - \sqrt{2} + \sqrt{3}$

(3) $4 - \sqrt{2} - \sqrt{3}$ (4) None of these

64. The standard deviation of a variate x is σ. The standard deviation of the variable $\frac{ax + b}{c}$; a, b, c are constants, is

(1) $\left(\frac{a}{c}\right)\sigma$ (2) $\left|\frac{a}{c}\right|\sigma$

(3) $\left(\frac{a^2}{c^2}\right)\sigma$ (4) None

65. Assume R and S are (non-empty) relations in a set A. Which of the following relation given below is false

(1) If R and S are transitive, then $R \cup S$ is transitive.

(2) If R and S are transitive, then $R \cap S$ is transitive.

(3) If R and S are symmetric, then $R \cup S$ is symmetric.

(4) If R and S are reflexive, Then $R \cap S$ is reflexive.

66. Given that f and g are continuous functions on [0 a] satisfying $f(a-x) = f(x)$ and $g(x)+g(a-x) = 2$.

Then $\int_0^a f(x)g(x)dx$ is equal to :

(1) $\int_0^a f(x)dx$ (2) $\int_0^a g(x)dx$

(3) 0 (4) None of these

67. If $c_1 = y = \frac{1}{1+x^2}$ and $c_2 = y = \frac{x^2}{2}$ be two curves lying in XY-plane, then

(1) area bounded by curve $y = \frac{1}{1+x^2}$ and $y = 0$ is $\frac{\pi}{2}$

(2) area bounded by c_1 and c_2 is $\frac{\pi}{2} - 1$

(3) area bounded by c_1 and c_2 is $1 - \frac{\pi}{2}$

(4) area bounded by curve $y = \frac{1}{1+x^2}$ and x-axis is $\frac{\pi}{2}$

68. Let f and g be functions from the interval $[0, \infty)$ to the interval $[0, \infty)$, f being an increasing and g being a decreasing function. If $f\{g(0)\} = 0$ then

(1) $f\{g(x)\} \geq f\{g(0)\}$

(2) $g\{f(x)\} \leq g\{f(0)\}$

(3) $f\{g(2)\} = 7$

(4) None of these

69. $\frac{dy}{dx} + y = 2e^{2x}$ then y is

(1) $ce^{-x} + \frac{2}{3}e^{2x}$

(2) $(1+x)e^{-x} + \frac{2}{3}e^{2x} + c$

(3) $ce^{-x} + \frac{2}{3}e^{2x} + c$

(4) $e^{-x} + \frac{2}{3}e^{2x} + c$

70. If $\begin{vmatrix} a+x & a-x & a-x \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix} = 0$ then x is

(1) 0, 2a (2) a, 2a

(3) 0, 3a (4) None of these

71. The equation of a circle with origin as centre and passing through the vertices of an equilateral triangle whose median is of length 3a is

(1) $x^2 + y^2 = 9a^2$ (2) $x^2 + y^2 = 16a^2$

(3) $x^2 + y^2 = 4a^2$ (4) $x^2 + y^2 = a^2$

72. If a positive integer n is divisible by 9, then the sum of the digits of n is divisible by 9. So which statement is it contrapositive.

(1) (sum of digits of n is divisible by 9) $\Rightarrow$ (n is divisible by 9)

(2) (sum of digits of n is not divisible by 9) $\Rightarrow$ (n is not divisible by 9)

(3) (sum of digits of n is divisible by 9) $\Rightarrow$ (n is divisible by 9)

(4) None of these

73. Fifteen coupons are numbered 1, 2 ..... 15, respectively. Seven coupons are selected at random one at a time with replacement. The probability that the largest number appearing on a selected coupon is 9, is

(1) $\left(\frac{9}{16}\right)^6$ (2) $\left(\frac{8}{15}\right)^7$

(3) $\left(\frac{3}{5}\right)^7$ (4) None of these

74. If $a \leq 0$ then roots of $x^2 - 2a|x-a| - 3a^2 = 0$ is

(1) $(-1+\sqrt{6})a$ (2) $(\sqrt{6}-1)a$

(3) a (4) None of these

75. If $n(A) = 1000, n(B) = 500$ and if $n(A \cap B) \geq 1$ and $n(A \cup B) = p$, then

(1) $500 \leq p \leq 1000$ (2) $1001 \leq p \leq 1498$

(3) $1000 \leq p \leq 1498$ (4) $1000 \leq p \leq 1499$

76. If $y = \log_2\{\log_2(x)\}$, then $\frac{dy}{dx}$ is

(1) $\frac{\log_2 e}{x \ln x}$ (2) $\frac{2.3026}{x \ln x \ln 2}$

(3) $\frac{1}{\ln(2x)^x}$ (4) None of these

77. $f(x) = \sin|x| . f(x)$ is not differentiable at

(1) $x = 0$ only (2) all x

(3) multiples of π (4) multiples of $\frac{\pi}{2}$

78. The angle between the pair of tangents drawn to the ellipse $3x^2 + 2y^2 = 5$ from the point (1,2) is

(1) $\tan^{-1}\left(\frac{12}{5}\right)$ (2) $\tan^{-1}\left(6\sqrt{5}\right)$

(3) $\tan^{-1}\left(\frac{12}{\sqrt{5}}\right)$ (4) $\tan^{-1}\left(12\sqrt{5}\right)$

79. The least integral value α of x such that $\frac{x-5}{x^2+5x-14} > 1$, satisfies :

(1) $\alpha^2 + 3\alpha - 4 = 0$
(2) $\alpha^2 - 5\alpha + 4 = 0$
(3) $\alpha^2 - 7\alpha + 6 = 0$
(4) $\alpha^2 + 5\alpha - 6 = 0$

80. The x satisfying $\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x$ are

(1) $1, 0$ (2) $1, -1$

(3) $0, \frac{1}{2}$ (4) None of these

Section - B

81. The degree of differential equation satisfying the relation $\sqrt{1+x^2} + \sqrt{1+y^2} = \lambda(x\sqrt{1+y^2} - y\sqrt{1+x^2})$ is _____ .

82. If the second term in the expansion $\left(\sqrt[13]{a} + \frac{a}{\sqrt{a^{-1}}}\right)^n$ is $14a^{5/2}$, then value of $\frac{{}^nC_3}{{}^nC_2}$ is _____ .

83. Given two independent events, if the probability that exactly one of them occurs is $\frac{26}{49}$ and the probability that none of them occurs is $\frac{15}{49}$, then the probability of more probable of the two events is _____ .

84. α, β be the roots of $x^2 - 3x + a = 0$ and γ, δ be the roots of $x^2 - 12x + b = 0$ and numbers $\alpha, \beta, \gamma, \delta$ (in order) form an increasing G.P. then value of $a + b$ is _____ .

85. The probability of A = Probability of B = Probability of $C = \frac{1}{4}$

$P(A) \cap P(B) \cap P(C) = 0$, $P(B \cap C) = 0$ and $P(A \cap C) = \frac{1}{8}$, $P(A \cap B) = 0$

the probability that atleast one of the events A, B, C exists is _____ .

86. If $\sum_{r=0}^{n} \frac{r+2}{r+1} {}^nC_r = \frac{2^8 - 1}{6}$, then n is _____ .

87. $f(x) = \frac{\sin 3x}{\sin x}$, when $x \neq 0$
$= k$, when $x = 0$
for the function to be continuous, k should be _____ .

88. If $\int_0^{\pi/3} \frac{\cos x + \sin x}{\sqrt{1 + \sin 2x}} dx = k\pi$ then value of k is _____ .

89. The S.D. of the following data is nearly

x_i	140	145	150	155	160	165	170	175
f_i	4	6	15	30	36	24	8	2

is _____ .

90. A line passes through (2,2) and is perpendicular to the line $3x + y = 3$, its y intercept is _____ .

Space for Rough Work